

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 3: Counting Life: Characteristics of Populations and Direct Counting in the School Compound

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	D	B	B	C	A	D	A	C	C

USE AFTER: PREDICT PHASE — Predict: recall the river video and write a prediction (fact or opinion, how to turn it into evidence)

Q1. Villagers beside a Kenyan river say, "There used to be many fish." Which action would best turn this statement into evidence?

- A. Look at the river and guess how many fish are there
- B. Count the fish in a known area at different times and compare the numbers**
- C. Ask more villagers whether they agree
- D. Wait to see if the smell of the water goes away

Correct answer: B. Evidence comes from measurement. Counting individuals in a known area at different times lets us compare numbers, whereas opinions and guesses are not reliable.

USE AFTER: PREDICT PHASE — Predict: introducing the word 'population' (What do we call all the fish of one species in the river?)

Q2. What do we call all the fish of one species living together in the same river?

- A. A community
- B. A habitat
- C. An ecosystem
- D. A population**

Correct answer: D. A population is all the individuals of one species living in the same place at the same time. A community includes all the species, and a habitat is the place where they live.

USE AFTER: OBSERVE PHASE — Observe: jigsaw counting at Site A and Site B, Calculator finds density

Q3. A group marks a 10 m x 10 m area on the school field and counts 250 grass tufts. What is the population density of grass tufts?

- A. 25 per m²
- B. 2.5 per m²**
- C. 0.4 per m²
- D. 350 per m²

Correct answer: B. Area = 10 m x 10 m = 100 m². Density = number of individuals / area = 250 / 100 = 2.5 per m².

USE AFTER: OBSERVE PHASE — Observe: teacher demonstration of marking the 10 m x 10 m area and agreeing on what counts as ONE plant

Q4. During the jigsaw counting, why must the group agree on what counts as ONE plant before they start counting?

- A. So that the density will always be a whole number
- B. So that everyone counts in the same way and no plant is counted twice or missed**
- C. So that the counting finishes faster
- D. So that the area marked will be larger

Correct answer: B. A clear counting rule keeps the count consistent and honest. It avoids double counting and missed plants, which would make the data unreliable.

USE AFTER: EXPLAIN PHASE — Explain: difference between population size and population density, and correcting the misconception that more individuals always means higher density

Q5. Site A has 300 weeds in a 10 m x 10 m area. Site B has 300 weeds in a 20 m x 10 m area. Which statement is correct?

- A. Both sites have the same density because they have the same number of weeds
- B. Site B has the higher density because its area is larger
- C. Site A has the higher density because the same number of weeds is found in a smaller area**
- D. Density cannot be compared unless the number of weeds is different

Correct answer: C. Site A density = $300 / 100 = 3$ per m^2 . Site B density = $300 / 200 = 1.5$ per m^2 . The same number in a smaller area means a higher density, so number alone does not show density.

USE AFTER: EXPLAIN PHASE — Explain: when direct counting is suitable and whether it would work for fish or algae in a river

Q6. Which situation is suitable for the direct counting method?

- A. Counting the trees growing in a small school plot**
- B. Counting the insects flying over a whole forest
- C. Counting all the fish in Lake Victoria
- D. Counting all the algae cells in a river

Correct answer: A. Direct counting suits small populations whose individuals are easy to see, identify and count, such as trees in a small plot. Fish, algae and flying insects are too many, hidden or mobile.

USE AFTER: EXPLAIN PHASE — Explain: probing the river link (If fish swim, hide in deep water and cannot be seen, can we count every one?)

Q7. Why would direct counting of every fish in a river be difficult?

- A. Rivers have no area that can be measured
- B. Fish do not belong to any population
- C. Fish density cannot be calculated using area
- D. Fish can move and hide in deep water, so they cannot all be seen and counted**

Correct answer: D. Fish swim, hide and are hidden in deep water, so counting every individual is not possible. Other methods, such as sampling, are needed.

USE AFTER: DQB PHASE — DQB: update the board and ask 'Did our two 10 m x 10 m areas tell us about the whole field?'

Q8. A group counted grass in two 10 m x 10 m areas of the school field. A learner asks, "Do these two counts tell us how the grass is spread over the whole field?" What is the best answer?

- A. Not necessarily, because two small areas may not represent the whole field, so we need to test if the distribution is uniform**
- B. No, because density is not affected by where the area is
- C. Yes, because any two areas will always show the whole field
- D. Yes, because direct counting always works for the whole field

Correct answer: A. Two small areas give information only about those places. Conditions such as shade, soil and trampling can vary, so we must investigate whether the grass is spread evenly across the field.

USE AFTER: END OF LESSON — Model: exit ticket question 2 (density for 180 plants in 100 m^2)

Q9. A site has 180 plants in a 100 m^2 area. What is the population density?

- A. 0.18 per m^2
- B. 18 per m^2
- C. 1.8 per m^2**
- D. 80 per m^2

Correct answer: C. Population density = number of individuals / area = $180 / 100 = 1.8$ per m^2 .

Q10. Which statement correctly defines population density?

- A. The total number of species in an ecosystem
- B. The number of individuals counted in the whole habitat
- C. The number of individuals of a species per unit area**
- D. The area occupied by one individual of a species only

Correct answer: C. Population density is the number of individuals per unit area, calculated as number of individuals divided by area sampled.